

EHSAN ULLAH

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EXPERIENCE

Metapolaris GmbH | *Software Engineer* 4/2025 – Present

- Shipped full-stack production features in TypeScript and Java, with a focus on scalability and robustness.
- Built Airflow/Prefect workflows for data sync and anomaly detection across 8+ systems, saving 3 h/week.
- Rolled out backup and recovery solutions for PostgreSQL and multi-terabyte ClickHouse production databases.
- Configured GitOps deployments with Argo CD, reducing manual deployment effort from 30 min to under 5 min.
- Monitored production services with Prometheus, Grafana, and Uptime Kuma for reliability and incident visibility.

Strategic Systems International | *Software Engineer* 7/2023 – 3/2024

- Delivered 5+ features across Node.js and Python microservices with full ownership from design to production.
- Implemented effective optimizations across 12+ legacy APIs, reducing average response times from 4s to under 0.2s.
- Developed platform's core MySQL reporting feature that analyzed 100M+ records for data-driven decision making.

Upwork | *Software Engineer* 8/2021 – 6/2023

- Developed TypeScript microservices and REST APIs supporting seamless frontend integration.
- Created custom middlewares using Express to handle intricate business logic and manage server-side operations.
- Established Apache Superset dashboards for 6+ KPIs, reducing manual reporting effort by 2 h/week.

RESEARCH EXPERIENCE

Humanoid Robotics Lab - University of Bonn | *Research Assistant* 5/2024 – Present

- Applied reinforcement learning algorithms and simulation-to-reality (sim2real) transfer processes for robotic systems
- Built Python-based experiment pipelines to train, evaluate, and compare control policies across simulated tasks.

DZNE - German Center for Neurodegenerative Diseases | *Research Assistant* 7/2024 – 12/2025

- Designed VR based motion-capture experiments and ML models to classify neurodegenerative patients.

SKILLS

Languages: Java, Python, TypeScript, C++, SQL.

Frameworks: React, Node.js, FastAPI, Spring Boot.

Developer Tools: Git, Linux, Cursor, Claude Code.

DevOps & Cloud: Docker, Argo CD, CI/CD, AWS (ECS, EC2, S3, SQS, SNS).

General: Microservices, Event-Driven Architecture, gRPC, RDBMS, NoSQL Databases, ORMs, PyTorch.

Spoken Languages: English (C1), German (B1).

EDUCATION

University of Bonn 2024 – Present

Master of Science in Computer Science

Bonn, Germany

National University of Computer and Emerging Sciences (NUCES)

2019 – 2023

Bachelor of Science in Computer Science (Magna Cum Laude)

Lahore, Pakistan

PERSONAL PROJECTS

LLM Controlled CoBots | *Robotics & LLMs*

- Built VLM-powered robot control for unknown object classification and sorting, achieving 90% success rate.

F1TENTH Autonomous Racing with Deep Reinforcement Learning (PPO) | *RL & Robotics*

- Sim-to-real PPO agent for F1TENTH autonomous racing with stable high-speed control on the physical car.

Multilingual Named Entity Role Classification | *NLP & Deep Learning*

- Developed a PyTorch based model for multi-label, multi-class NER role classification on multilingual news articles.